

Internship Semester Report

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I would like to show my appreciation for all the persons who gave me a chance to complete my internship semester in Hochschule Hamm-Lippstadt (Department Lippstadt 1). I would specifically thank my internship supervisor Prof. Dr. Ulrich Schneider who guided me through weekly meetings and his professional advice in realizing my internship tasks. Prof. Dr. Ulrich Schneider also nurtured me in developing calculative approach which would enhance my future professional career as an Electronic Engineer.

I would also like to show my gratitude towards Herr Marc Ebmeyer for mentoring me on several design aspects of my internship tasks. His advices aided me in achieving several precision engineering requirements.

Abstract

The future career prospects in the field of electronics and information technology developed an interest in me to pursue a career in this field. The exponential developments in the field of electronics creates a requirement of hands on experience for an aspiring electronic engineer. Hence, I chose to participate in the internship opportunity at Hochschule Hamm-Lippstadt with Prof.Dr. Ulrich Schneider as my supervisor. The tasks offered to me of working with Topcon Total Robotic Robotic station PS-103A and Solidworks designing helped in acquiring relevant skills.

This paper is an internship report that gives a detailed information of the tasks and activities conducted as part of the internship. Project 1 includes introduction, tasks, hardware and software requirements, theoretical foundations, solution of tasks, results and implementations.

Project 2 includes introduction, tasks, hardware and software requirements, solutions of task and implementation results.

This paper also includes the conclusion and key lessons learned, declaration of originality and Bibliography.

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Chapter 1

PROJECT 1 - 3D POSITIONING WITH TOPCON ROBOTIC TOTAL STATION PS-103A

1.1 Introduction

The 3 dimensional Topcon Robotic Total Station PS-103 A is used for tracking the glass prism in spatial coordinate system. The official software available from the company to use the tracking functionality is costly and is only available on subscription basis. Hence, this created a need to develop a software for functions such as prism tracking, transmitting the tracking data to a Personal Computer and further data analysis in Matlab software. The user experience is kept in mind as the prism would be mounted on a remote control car and used for path tracking. The Tasks for this project are as follows:

1.2 Tasks for Project 1

1. Familiarization with existing system and Hardware setup.

This task consists of understanding all the functionalities of the Topcon system related to the project such as prism tracking, data sending protocols and coordinate positioning of the Topcon device.

2. Transmitting the measurement data over serial interface

Choose a suitable transmission protocol for the measurement data of tracking the prism. The protocol has to be secured, reliable and user friendly.

3. Receive the data via serial interface in Matlab

Write a matlab program for efficient data receiving. Determine the path and 3D position of the prism in a cartesian coordinate system. Also note the timing information for the measured data packet

4. System testing and data analysis

Use mathematical analysis tools for further analysis of the data to get various other information like velocity and acceleration in respective axis. Present the car data in a visual manner with the help of matlab.

5. Optimization

Optimize the system for better user experience, reduce latency and other complexities.

6. Documentation in the HSHL Wiki

Documentation of the above finding and developments for future use.

1.3 Hardware Requirements

To realize this project the following hardware equipments and components are required:

1. Topcon Robotic Total Station PS-103A



Figure 1.1: Topcon Robotic Total Station PS-103A

The Topcon Robotic Total Station PS-103A shown in figure 1.1 has been used for the measurements and tracking of the prism. [1] This device is sold by the company Topcon Positioning Company, Inc. [2] An entire kit including power supplies is available as a package product.

2. Topcon ATP1 360 Prism

The prism shown in 1.2 has been used for mounting on a tracked device is sold by Topcon Positioning Company [3]

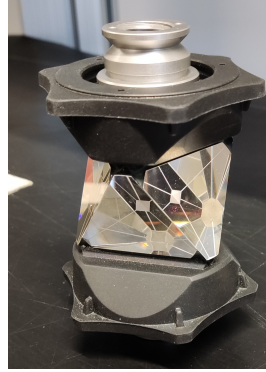


Figure 1.2: Prism

3. Personal Computer- Bluetooth

A personal computer or a Laptop with Bluetooth communication protocol is needed for further measurement data usage. A bluetooth dongle was used in this project for providing connectivity to the personal computer.

1.4 Software Requirements

1. Matlab software

The matlab software is used for high level data processing and other functions. Any version of matlab can be used.

2. Windows 10

Windows 10 system was used for ideal file storage.

3. SVN - Version Management System.

SVN version control was used for version management.

1.5 Theoretical Foundations

Cosine Theorem

The topcon device needs to be placed at exactly the same spot every time for usage to orient itself perfectly with the movement of the prism. This cosine theorem requirement is not need for the initially mentioned above tasks but is used for enhancing the user experience.

During recording of the RC car track two coordinate points are used for fixing the

orientation of the track. The y - axis of the initial point A is set to 0 units. The x coordinate of the A point can vary. The B coordinates are then measured and the data is saved. It is made sure that both A and B points are location on the track and are in the linear section of the track. This gives us a fixed a fixed axis that is y axis and the distance between A and B points.

The same process is done to measure the A and B points when only tracking the car is need. The Matlab code calculates the angle for the reorientation and reorients the center on the graph and the coordinate system. The reoriented coordinates are presented. [4]

Kalman filter

The Kalman filter is used for calculating the velocity and acceleration of the prism in Cartesian coordinate system axis respectively.[5] The main algorithm code for matlab of Kalman filter was provided by Prof.Schneider Ulrich.

1.6 Solutions of Tasks

This section gives the solution of each task.

Task 1 - Familiarization with existing system and Hardware setup solution:

The solution to this task involves understanding the functionality of the Topcon station. The setting of the Topcon station includes exploring the topcon communication protocols such as Bluetooth system and RS-232 serial protocol for the transferring the X , Y and Z data to the personal computer. Setting up the topcon station for tracking the prism. Also setting up the motor movement of the total station for prism tracking. The coordinate positioning was understood form prism location. The Topcon station datasheet was used extensively for this task. [1] Figure 1.3 shows various steps performed during Task 1.

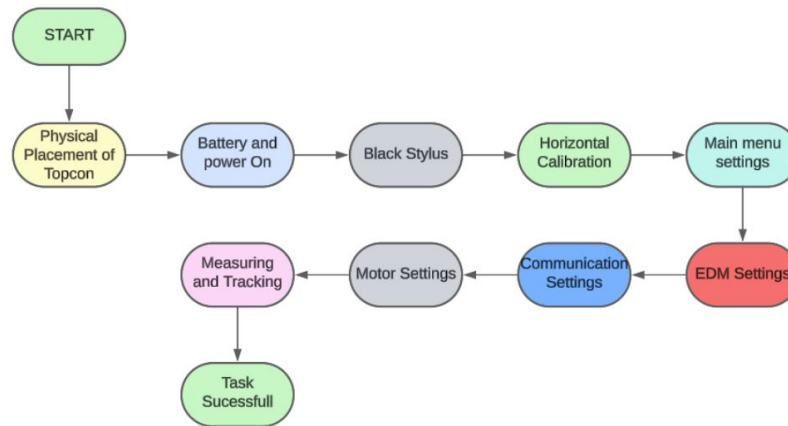


Figure 1.3: Tasks done for familiarization

Task 2 - Transmitting the measurement data over serial interface solution:

There are two main serial data transmission protocols available in the topcon total station that are user friendly. The RS- 232 protocol and the bluetooth protocol. The Bluetooth protocol was chosen as it is wireless and user friendly.

The Figure 1.4 shows the setup of the bluetooth protocol in topcon station. All the bluetooth settings are chosen as per the datasheet instructions.

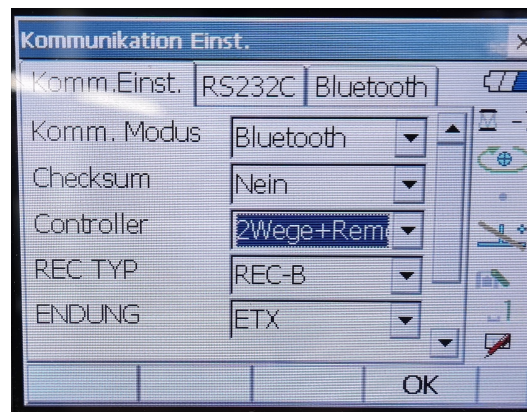


Figure 1.4: Bluetooth protocol Setup

Task 3 - Receive the data via serial interface in Matlab solution:

A matlab program was written to receive the data in the personal computer. The matlab 2019 version uses serial communication ports library while all the versions from 2020 onward use the bluetooth protocol directly. Hence the matlab code that

was developed check the current matlab version in use and uses the respective library to communicate and receive data.

```
% Set up serial port communication
versionInfo = ver('MATLAB'); % Detailed version info
bluetoothSupported = contains(versionInfo.Release, {'R2020b', 'R2021', 'R2022', 'R2023', 'R2024'});
if bluetoothSupported
    disp('Bluetooth is supported in this MATLAB version. ');
    % Attempt Bluetooth communication
    devices = bluetoothlist;
    disp(devices);
    s = bluetooth("PS1031404", 1); % Channel 1 is typical
else
    s = serialport("COM4", 9600); % Adjust COM port and baud rate as needed
    % Inform the user the terminal is running
    disp('Starting continuous serial terminal. ');
end
```

Figure 1.5: Bluetooth communication code in matlab

The figure 1.5 show the code that was used establishing the bluetooth communication. It checks the version of matlab first using the `ver()` function , if the bluetooth protocol is supported then it finds out devices using the "bluetoothlist" command . Then it checks the topcon station device using the given PS1031404 name and channel 1. If these conditions are true then the " if statement" becomes true or if it is false then the compiler goes to the else statement where it established the serial communication via the personal computer operating system using the connected communication port and the baud rate. The code was created with the aid of chatgpt.

1.7 Results of implementations

Task 4 - System testing and data analysis:

The further tasks are entwined with results and implementations as they include an integrated process of analyzing and optimizing the measured data simultaneously.

```
z = data(i-1);
x = data(i-2)+2*refX ; % x coordinate
y = data(i-3) ; % y coordinate
collectedData{end+1, 1} = x;
collectedData{end, 2} = y;
collectedData{end, 3} = z;
collectedData{end, 4} = char(timestamp); % Convert timestamp to string
```

Figure 1.6: Timestamp for each data packet

The incoming data was then added with time stamps the moment the data was received in matlab. The `char(timestamp)` command is used as shown in figure 1.7. This was achieved with the aid of Chatgpt.

X-Coordinate	Y-Coordinate	Z-Coordinate	Timestamp
0.770000	-0.480000	1.820000	2025-01-10 10:04:07.532
0.770000	-0.480000	1.820000	2025-01-10 10:04:07.660
0.770000	-0.480000	1.820000	2025-01-10 10:04:07.826
0.770000	-0.480000	1.820000	2025-01-10 10:04:07.973
0.770000	-0.480000	1.820000	2025-01-10 10:04:08.143
0.770000	-0.480000	1.820000	2025-01-10 10:04:08.300
0.770000	-0.480000	1.820000	2025-01-10 10:04:08.455
0.770000	-0.480000	1.820000	2025-01-10 10:04:08.607
0.770000	-0.480000	1.820000	2025-01-10 10:04:08.781
0.770000	-0.480000	1.820000	2025-01-10 10:04:08.925
0.770000	-0.480000	1.820000	2025-01-10 10:04:09.090
0.770000	-0.480000	1.820000	2025-01-10 10:04:09.247
0.770000	-0.480000	1.820000	2025-01-10 10:04:09.409
0.770000	-0.480000	1.820000	2025-01-10 10:04:09.566
0.770000	-0.480000	1.820000	2025-01-10 10:04:09.721
0.770000	-0.480000	1.820000	2025-01-10 10:04:09.909

Figure 1.7: Implementation of Bluetooth protocol

The figure 1.6 shows the data transmitted via bluetooth to the matlab software. It shows the X,Y , Z coordinates and timestamp of the prism. The data is transmitted with the **5 Hz frequency**. The kalman filter was used for data analysis here as it

```
% File path of the external file
filename = 'collected_data_with_timestamps.txt'; % Ensure the file exists in the same directory

% Read raw file content for debugging
rawData = fileread(filename);
disp('Raw File Content:');
disp(rawData);

% Read the file with explicit import options
opts = detectImportOptions(filename, 'Delimiter', '\t\t'); % Two tabs as delimiter
opts.VariableNames = {'x', 'y', 'z', 'timestamp'};
opts.VariableTypes = {'double', 'double', 'double', 'string'}; % Explicit data types

% Load the data
inputData = readtable(filename, opts);

% Remove rows with missing data
inputData = rmmissing(inputData);

% Clean and convert timestamps
try
    inputData.timestamp = datetime(inputData.timestamp, 'InputFormat', 'yyyy-MM-dd HH:mm:ss.SSS');
catch ME
    error('Timestamp conversion failed. Check the input file formatting.');
```

```
end

% Extract x, y, and timestamp data
x_positions = inputData.x; % Numeric x-coordinates
y_positions = inputData.y; % Numeric y-coordinates
timestamps = inputData.timestamp; % Datetime array
```

Figure 1.8: Additions to Kalman Filter

provide the velocity and acceleration with respect to the X and Y axis. The initial code for the kalman filter was provided by Prof. Schneider Ulrich. The Figure 1.9 shows the additions being realized with the help of Chatgpt to give the coordinate and time stamp data input.

The results of the kalman filter shown in Figure 1.8 are the x position versus time ,y position versus time , x velocity versus time, y velocity versus time ,x acceleration

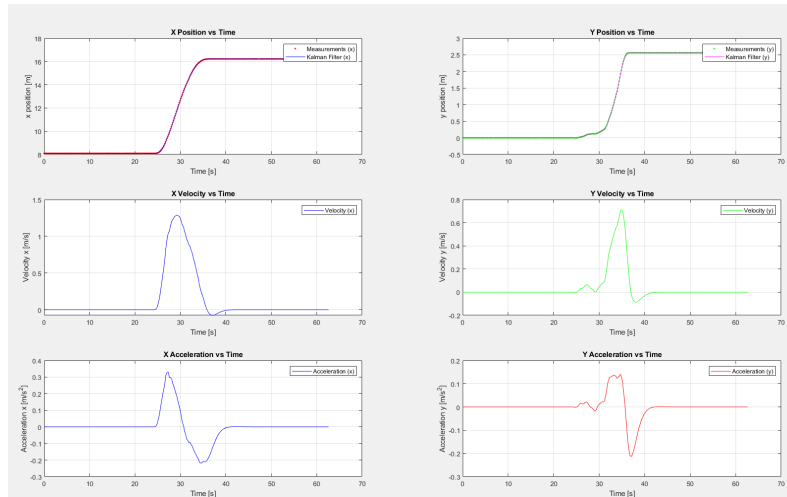


Figure 1.9: Kalman filter results

versus time and y acceleration versus time graphs.

Task 5 Optimization

Several optimizations were carried out for better user experience and accuracy. Initially plan was to record the track using the prism and the use normal vectors to determine the points that would create the side tracks of the path. The results are as follows.

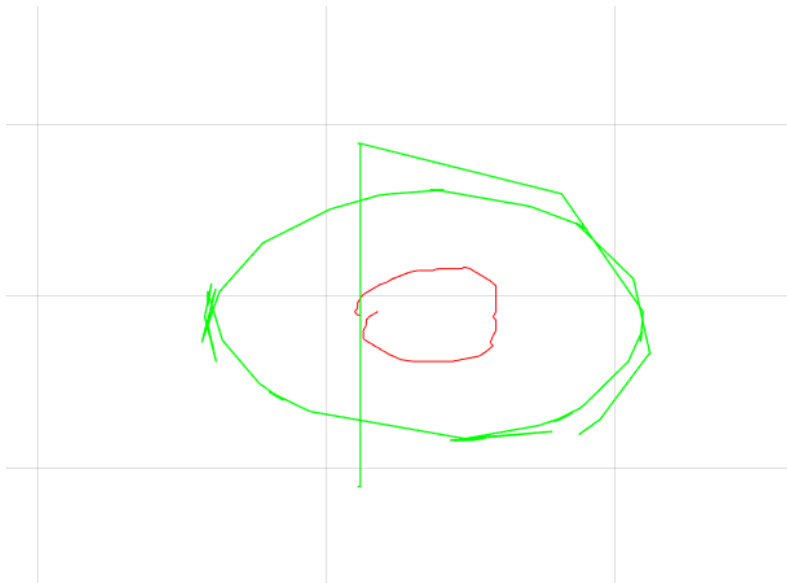


Figure 1.10: Normal vector interpolation

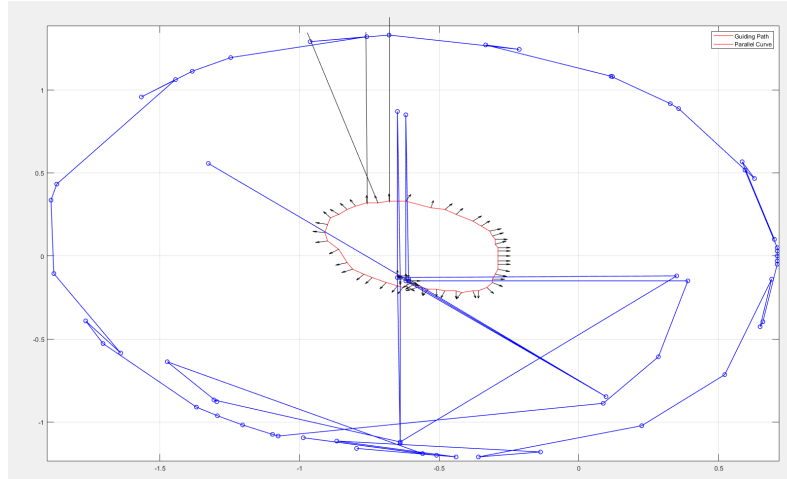


Figure 1.11: Cubic Spline Interpolation

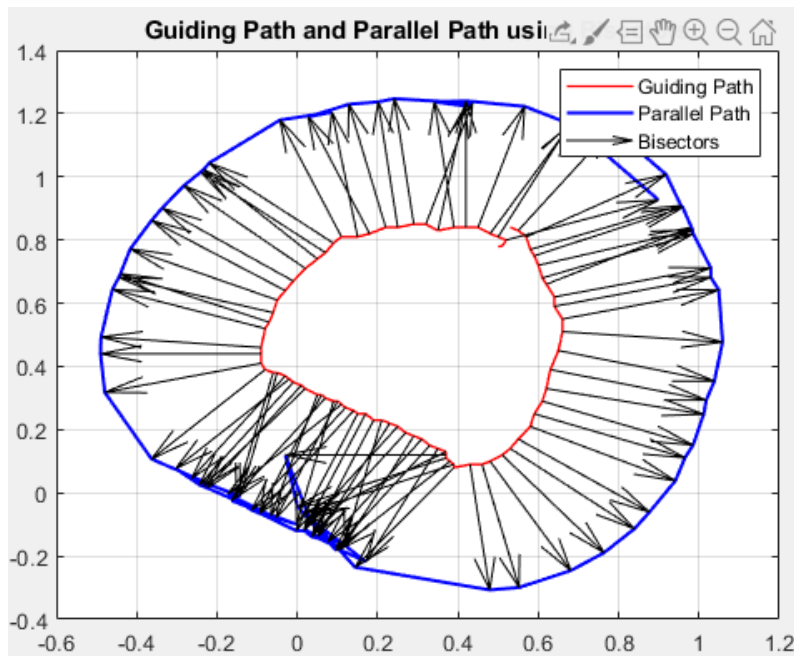


Figure 1.12: Polynomial interpolation

As it can be seen from the figure 1.10, 1.11 and 1.12 that the methods of polynomial interpolation, cubic spline interpolation and normal vector interpolation were not creating the ideal path. This was because the normal vectors would overlap each other. This was caused by the the magnitude of the normal vectors was getting affected by the the curvature of the path. To calculate the curvature at every single point and recalculating the magnitude of every vector is a tedious method. This would also increase the code complexity and time complexity.

This would slow down the whole processing in matlab. Hence a different method was applied all together. The previous algebraic method was discarded and a geometric method was adopted. This geometric method would calculate the angle bisectors at every point inbetween the line and the would determine the points of the side tracks of the track.

This also requires to give in the required track width and start and end points. This actually helped achieve a more customized solution as you can easily adjust the width of he track, start and end points at the start of the data processing in matlab as shown in figure 1.13. The figure 1.14 shows the full optimal track in red color and

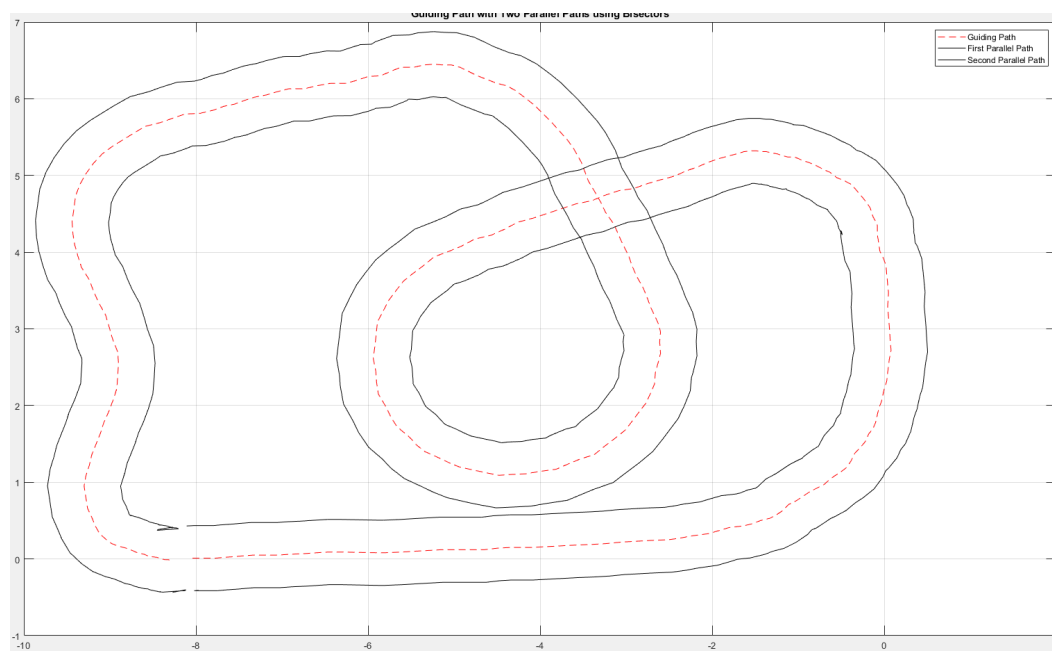


Figure 1.13: Full track after optimization

the path of the car in black color including the reorientation with the help of cosine theorem.

Task 6 Documentation in the HSHL Wiki

A documentation was created in HSHL wiki as a detailed guide with image and illustrations for setting up and usage of the Topcon station system. This helps future users of this system. [6]

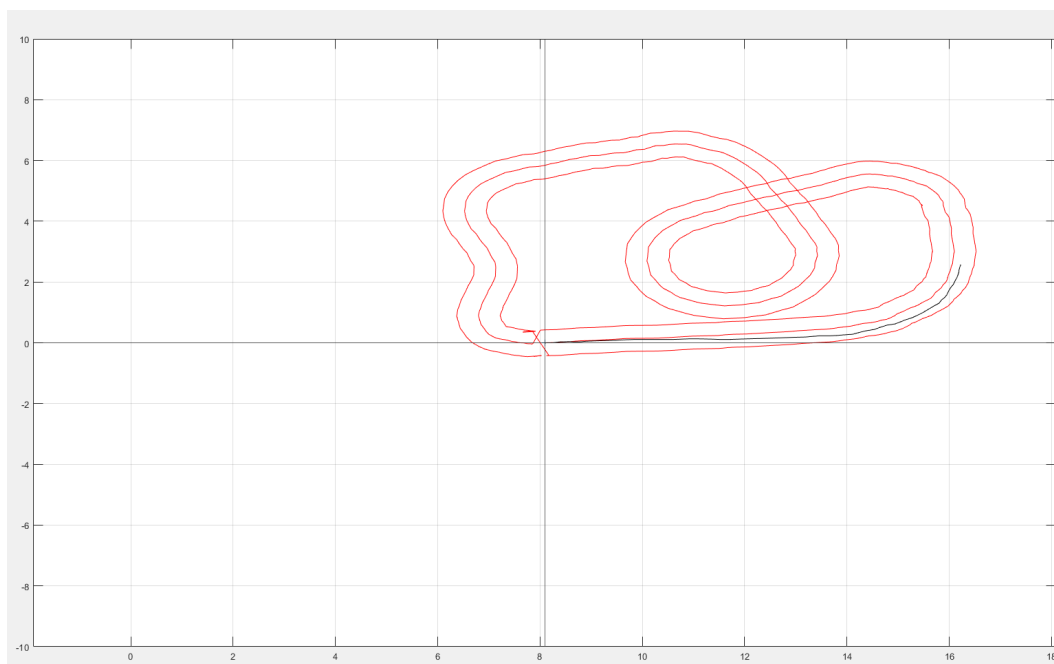


Figure 1.14: Full track with car path

Chapter 2

PROJECT 2 - RC CAR RACK IN SOLIDWORKS

2.1 Introduction

Testing RC cars is needed when developing several software features of the car. Hence there was a need to develop an RC car rack so that the RC car can be tested with ease.

2.2 Task for project 2

Create a rack for the JetRacer -02 car so that it can be tested easily. The rack should lift the vehicle so that the wheels do not touch the ground. Design the part in SolidWorks and then 3D print the rack.

2.3 Hardware Requirements

1. 3D printer needed for printing the rack which was provided by the Hochschule Hamm-Lippstadt University Lab.
2. Personal computer needed for SolidWorks designing.

2.4 Software Requirements

SolidWorks 2019 needed for designing the 3D part.

2.5 Solution of the Task

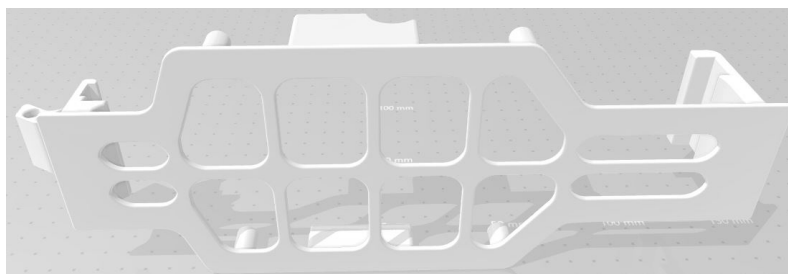


Figure 2.1: Solidworks Implementation Side View

This section shows the SolidWorks designing of the rack from top view and side view in figure 2.1 and 2.1.

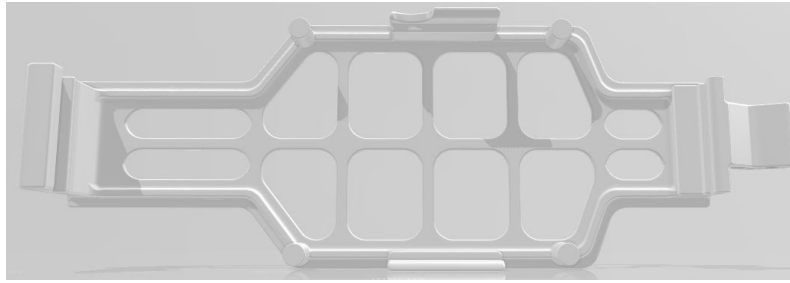


Figure 2.2: Solidworks Implementation Top View

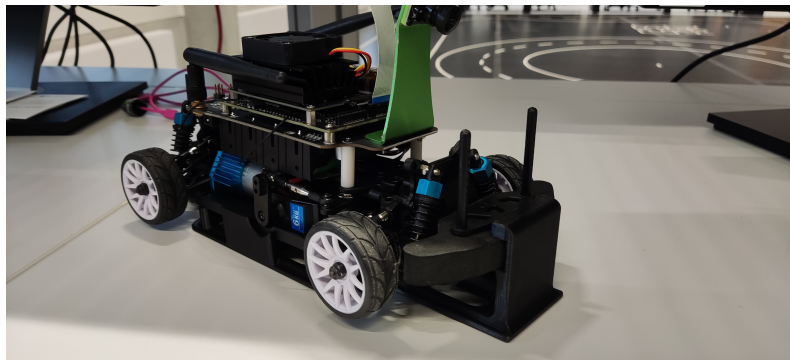


Figure 2.3: Perfect fit for the car

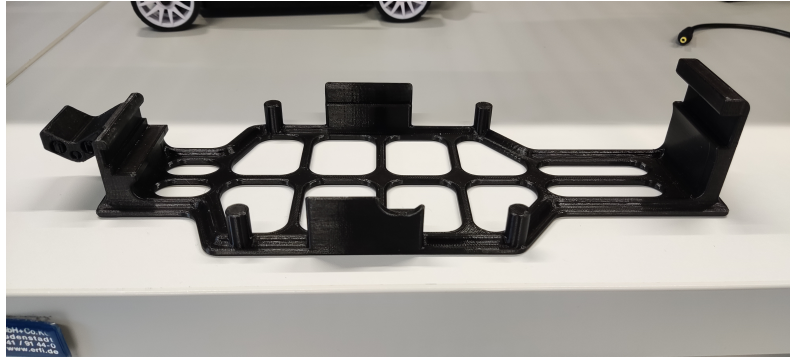


Figure 2.4: Final Product

2.6 Results and Implementations

The implementation results in figure 2.3, 2.4 and 2.5 show a rack fits perfectly onto the car. The wheels of the car are just above the ground and does not touch the floor. Hence the functionality is achieved. The back end of the car has complex designed part that required a precision engineering of upto 1 millimeter to have a perfect fit.

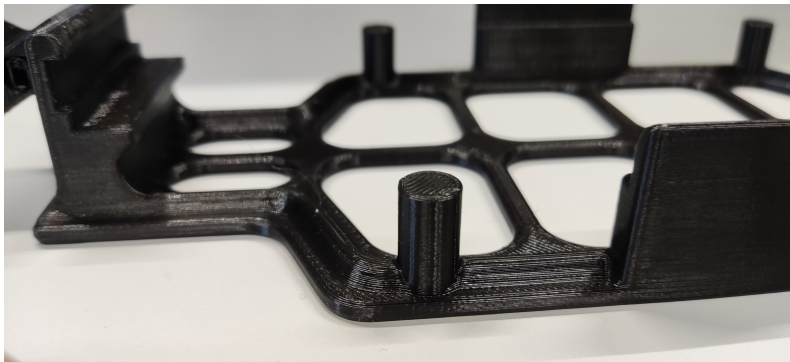


Figure 2.5: Precision Engineering at the back

Chapter 3

CONCLUSION

To conclude, every single aspect of the internship progress has been documented in this report. I have explored and learnt various things from this internship. The internship experience deepens my understanding about concepts of related to software development in matlab, use of Computer aided design (CAD) software solidworks and the operation of Topcon Robotic Station and its Bluetooth protocol. The project also helped me develop a global view of working with electronic systems and can prove to be crucial for my further career.

3.1 Key lessons learned

- Matlab programming language and its usage for data processing.
- Solidworks design for designing precision engineering applications
- Working under professional guidance of the supervisor.
- Time management and weekly meeting update with Prof. Schneider Ulrich.
- SVN version control and management.
- Data manipulation and other mathematical concepts such as cosine theorem and Kalman filter.

Chapter 4

DECLARATION OF ORIGINALITY

I, Yashodhan Vishvesh Deshpande, herewith declare that I have composed this internship report by myself and without the use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The paper and work in the same or similar form have not been submitted to any examination body and have not been published. This paper was not yet, even in part, used in another examination or as a course performance. I agree that my work may be checked by a plagiarism checker.

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